CSE350: Digital Electronics & Pulse Techniques

Lecture 01: Course Introduction & CSE250 Review

Prepared by Ihzaz Abrar Sadik[ZAZ]

Marks Distribution

Assessment Tools	Weightage (%)
Assignment	10
Quizzes	20
Midterm Exam	20
Final Exam	30
Lab work	20

Course Content

70%	Digital Electronics	Digital Logic Families
		DC Analysis
30%	Pulse Techniques	Signal Generators
		A/D, D/A Converters

CSE250 Review

Charge: *A fundamental & characteristic property of the basic elements of an atom.*

🔷The charge of a proton, an electron & a neutron is respectively positive, negative & zero. 🔷Coulomb's law is used to calculate the force between charged particles.

Voltage Difference: *The amount of work needed to transfer an unit positive charge from one point to another.*

If 'q' coulomb charge is used instead from point 'B' to 'A'', $V_A - V_B = \left(\frac{W}{q}\right)$

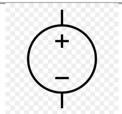
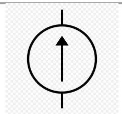
Current: *The rate of flow of charge through any element is called current.*

$$I = \frac{dq}{dt}$$

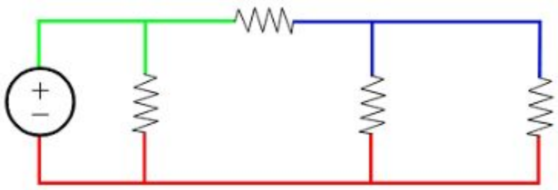
Ohm's Law: *At a fixed temperature, the amount of current flowing through a conductor/element is directly proportional to the voltage difference across the conductor/element.*

🔷$I \propto V$ □ $I = GV = V/R$; Where, 'G' is the conductance & 'R' = 1/G is the resistance of the conductor/element.

Some Important Electrical Components:

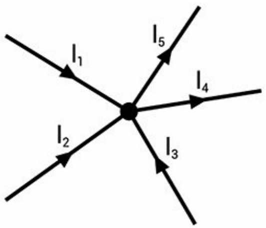
Name	Symbol	Characteristics
Voltage Source		Constant output voltage (unknown current)
Current Source		Constant output current (unknown voltage)
Capacitor		$C = Q/V$

Node: *A node is a point in a circuit which may connect one or more elements.* The number of nodes in a circuit is equal to the number of unique voltage points.



Here, the same colored points are effectively a single node & have the same voltage

KCL: *The algebraic sum of currents entering a node is equal to the sum of currents leaving the node.*



$$I_1 + I_2 + I_3 = I_4 + I_5$$

Mesh: *A loop that does not contain any other loops within it; i.e. a closed path in which no component or segment is encountered more than once.*

In the circuit at the right, there are total 6 loops which can be written as follows (order doesn't matter):

Loop 1 consists of 10 V, 5 , and 2 . *(Mesh)*

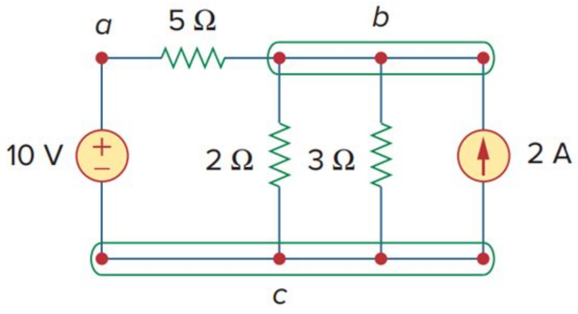
Loop 2: 10 V, 5 , and 3 .

Loop 3: 10 V, 5 , and 2 A.

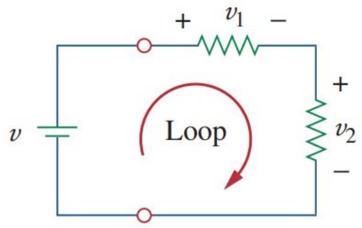
Loop 4: 2 and 3 . *(Mesh)*

Loop 5: 2 and 2 A.

Loop 6: 3 and 2 A. *(Mesh)*



KVL: *The algebraic sum of voltages across all the elements of a mesh is zero.*

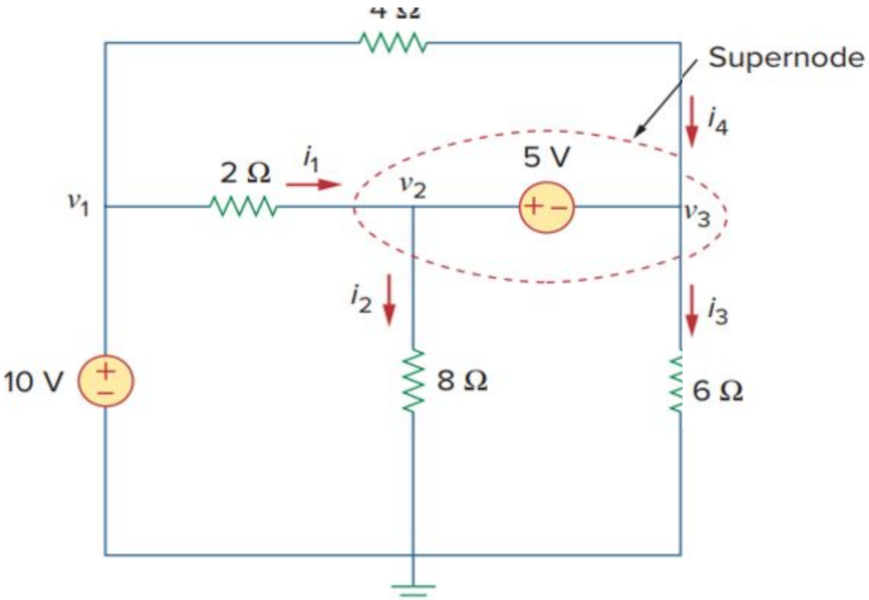


$$KVL: -v + v_1 + v_2 = 0$$

Nodal Analysis: *A process of finding node voltage values w.r.t. a reference node, using KCL.*

- **Step1:** Mark all the unique nodes and name them.
- **Step2:** Look for supernodes (combination of two nodes with a voltage source between them)
- **Step3:** Define a reference node.
- **Step4:** Apply KCL and find equations of nodes other than supernodes.
- **Step5:** Write the voltage relation of nodes forming supernodes.
- **Step6:** Combinedly apply KCL to supernode & find its equation.
- **Step7:** Solve the equations

Example



$$\frac{17V_1}{20} - \frac{V_2}{2} - \frac{V_3}{10} = 0$$

$$\frac{3V_1}{5} - \frac{3V_2}{2} - \frac{3V_3}{10} + 8 = 0$$

$$V_2 - V_3 = 5$$